GRADUATION PROJECT · COMPETITIVE ANALYSIS

Eshara vs. the Market

How Eshara compares with sign-language apps on Google Play and the App Store

A feature-by-feature comparison against 8 published apps, covering translation and AI, the avatar, learning, streaks and competition, the Kids section, chat and live conversation — and what makes Eshara different.

PLATFORM
AI-Powered Arabic Sign Language Accessibility Platform

INSTITUTION
Egyptian E-Learning University (EELU)

DATE
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THE GAP

Most sign-language apps are not built for us

The best-known apps are English-based, and the Arabic apps on the stores cover only one piece of the problem.

What is on the stores today

- **Hand Talk, Lingvano, PopSign, Sign Language AI** — English-based: ASL, BSL, Libras. No Arabic, no Egyptian Sign Language.
- **Al-Turjuman** — Arabic, but one direction only: speech/text → sign. It cannot read a Deaf person's signs.
- **Arabic SL App for the Deaf** — a static learning library: no AI, no translation.
- **BrightSign** — needs an expensive smart glove. **Ava** — captions only, no sign language.
- No app combines translation, learning, gamification, kids and community.

What Eshara brings

- Arabic first, starting with **Egyptian Sign Language** and Egyptian Arabic voice.
- **Two-way live translation** with the phone camera only — sign → text → speech and voice → text → sign.
- A **3D avatar** tied to translation, dictionary and lessons.
- **Learning** with camera practice, plus **streak, XP, daily challenges and a leaderboard**.
- A safe **Kids mode**, **chat and calls** with live captions, and an offline **emergency** screen.

APPS COMPARED

8 published apps

Every app below is available on Google Play and/or the App Store. Details come from the store listings and the Eshara survey.

App	Platforms	Sign language / language	What it does	Main gap vs. Eshara
Hand Talk	Android · iOS	ASL, BSL, Libras — English / Portuguese	Translates text and audio into sign language through 3D avatars (Hugo, Maya); rate and save translations, adjust speed, customize clothes and background.	No Arabic. One direction only — no camera recognition.
Al-Turjuman	Android · iOS	Arabic Sign Language	Translates written and spoken Arabic into Arabic Sign Language with 3D characters; 4,000+ word dictionary; offline; share as a video on WhatsApp.	One direction only. No learning path or gamification.
Sign Language AI	Android	ASL — English	Camera recognizes 2,600+ signs and the alphabet; English text or voice to sign; dictionary with categories and handshape search.	No Arabic. No avatar, lessons or community.
Lingvano	Android · iOS · Web	ASL, BSL, ÖGS	Bite-sized lessons, vocabulary trainer, dictionary, sign mirror; streaks, awards and streak freezes; reminders.	No Arabic. Learning only — no translation.
PopSign	Android · iOS	ASL	Bubble-popping game that checks isolated signs with the front camera; works offline.	No Arabic. Isolated words only.
BrightSign	Android · iOS (companion)	Customizable; Arabic text/voice output	Smart glove translates signs into text and speech; the other side replies by voice.	Needs expensive external hardware.
Ava	Android · iOS · Web · Desktop	None	Live speech-to-text captions grouped by speaker.	No sign language at all.
Arabic SL App for the Deaf	Android	Arabic Sign Language	Offline, ad-supported library: alphabet, numbers, days of the week and daily signs.	Learning library only — no AI, no translation.

COMPARISON 1 OF 3

Translation, Avatar & AI

Eshara is the only app that translates both ways in Arabic and Egyptian Sign Language using just the phone camera.

Translation & AI	Eshara	Hand Talk	Al-Turjuman	Sign Language AI	Lingvano	PopSign	BrightSign	Ava	Arabic SL App for the Deaf
Arabic Sign Language	✓	✗	✓	✗	✗	✗	~	✗	✓
Egyptian Sign Language & Egyptian Arabic voice	✓	✗	✗	✗	✗	✗	✗	✗	✗
Sign → Text with the phone camera	✓	✗	✗	✓	~	~	✗	✗	✗
Sign → Speech	✓	✗	✗	✗	✗	✗	✓	✗	✗
Text → Sign	✓	✓	✓	✓	✗	✗	✗	✗	✗
Voice → Sign	✓	✓	✓	✓	✗	✗	✗	✗	✗
Two-way live conversation in one app	✓	✗	✗	~	✗	✗	✓	✗	✗
3D avatar	✓	✓	✓	✗	✗	✗	✗	✗	✗
Recognition from letters up to continuous sentences	✓	✗	✗	~	✗	~	~	✗	✗
Hand + body + face (non-manual signs)	✓	✗	✗	✗	✗	✗	✗	✗	✗
Fingerspelling fallback for unmapped words	✓	✗	✗	✗	✗	✗	✗	✗	✗
No extra hardware (phone camera only)	✓	✓	✓	✓	✓	✓	✗	✓	✓
Works offline	~	✗	✓	✗	✗	✓	✓	✗	✓

✓ Available ~ Partial ✗ Not available, or not mentioned in the app's store listing Eshara column = features specified in the Eshara requirements

12 / 13
features covered by Eshara

6 / 13
best competitor (Al-Turjuman)

4
features only Eshara offers

COMPARISON 2 OF 3

Learning, Streak, Competition & Kids

Lingvano is the strongest learning app — but only for ASL / BSL. Eshara brings a full learning and gamification loop to Arabic, plus a safe space for children.

Learning, Gamification & Kids	Eshara	Hand Talk	Al-Turjuman	Sign Language AI	Lingvano	PopSign	BrightSign	Ava	Arabic SL App for the Deaf
Sign dictionary	✓	~	✓	✓	✓	✗	✗	✗	✓
Structured lessons by category	✓	✗	✗	✗	✓	✗	✗	✗	~
Camera practice with correct / try-again feedback	✓	✗	✗	✗	✓	✓	✗	✗	✗
Learning-progress tracking	✓	✗	✗	✗	✓	~	✗	✗	✗
Daily streak	✓	✗	✗	✗	✓	✗	✗	✗	✗
XP / badges	✓	✗	✗	✗	✓	~	✗	✗	✗
Daily challenges	✓	✗	✗	✗	✗	✗	✗	✗	✗
Opt-in leaderboard with nicknames	✓	✗	✗	✗	✗	✗	✗	✗	✗
Mini games	✓	✗	✗	✗	✗	✓	✗	✗	✗
Safe Kids mode (no chat or leaderboard)	✓	✗	✗	✗	✗	✗	✗	✗	✗

✓ Available ~ Partial ✗ Not available, or not mentioned in the app's store listing Eshara column = features specified in the Eshara requirements

10 / 10
features covered by Eshara

6 / 10
best competitor (Lingvano)

3
features only Eshara offers

COMPARISON 3 OF 3

Chat, Live Communication & Services

Competitors are single-purpose tools. Eshara is a communication platform with chat, calls, live captions and an offline emergency screen.

Communication, Services & Platform	Eshara	Hand Talk	Al-Turjuman	Sign Language AI	Lingvano	PopSign	BrightSign	Ava	Arabic SL App for the Deaf
One-to-one chat between users	✓	✗	✗	✗	✗	✗	✗	✗	✗
Audio & video calls	✓	✗	✗	✗	✗	✗	✗	✗	✗
Live captions / sign translation during calls	✓	✗	✗	✗	✗	✗	✗	~	✗
Push notifications & learning reminders	✓	✗	✗	✗	✓	✗	✗	✗	✗
Offline emergency screen (Egyptian numbers, responder cards)	✓	✗	✗	✗	✗	✗	✗	✗	✗
Translation history / favorites	✓	✓	✗	✗	✗	✗	✗	✗	✗
Share a translation	✓	✗	✓	✗	✗	✗	✗	✗	✗
Avatar customization	✓	✓	✗	✗	✗	✗	✗	✗	✗
Playback speed control	✓	✓	✗	✗	✗	✗	✗	✗	✗
Mobile app + web app	✓	~	~	~	✓	~	~	✓	~
Arabic (RTL) + English interface	✓	✗	✓	✗	✗	✗	✗	✗	✓

✓ Available ~ Partial ✗ Not available, or not mentioned in the app's store listing Eshara column = features specified in the Eshara requirements

11 / 11
features covered by Eshara

3 / 11
best competitor (Hand Talk)

4
features only Eshara offers

WHAT MAKES ESHARA DIFFERENT

12 reasons Eshara stands out

01

Arabic & Egyptian first

Built for Arabic Sign Language starting with Egyptian Sign Language, Egyptian Arabic voice input and Egyptian emergency numbers. Hand Talk, Lingvano, PopSign and Sign Language AI are English-based (ASL / BSL / Libras).

02

True two-way translation

Sign → Text → Speech and Voice → Text → Sign in the same app, plus a split-screen Live Conversation Mode on one phone. Al-Turjuman and Hand Talk only go from speech/text to sign.

03

Phone camera only

On-device MediaPipe landmarks for hands, body and face — no gloves, no depth sensors, no multi-camera rigs (unlike BrightSign).

04

Avatar connected to AI & learning

3D avatar (neutral, male, female) used in translation, dictionary and lessons, with fingerspelling when a word has no mapped sign.

05

Learn by doing

Lessons, dictionary and camera practice where the recognizer tells the learner correct / try again — in Arabic, on mobile.

06

Streak & competition

XP, daily streak, daily challenges and an opt-in leaderboard with nicknames. Points are awarded by the server from verifiable events, so they cannot be faked.

07

Safe Kids mode

Mini games and learning for children with no chat and no leaderboard — none of the compared apps offers a dedicated kids space.

08

A community, not just a tool

One-to-one chat, audio/video calls and live captions or sign translation during calls, with push notifications and reminders.

09

Emergency assistance

Offline emergency screen with 122 / 123 / 180, large visual signs and full-screen cards to show a responder, one-tap call and SMS with location.

10

Privacy & security by design

Raw video never leaves the device — only landmarks are sent. HTTPS/WSS, JWT, ownership checks, login-attempt blocking and history saved only by user choice.

11

Accessible everywhere

Android, iOS and web; Arabic RTL and English LTR; WCAG 2.1 AA target, TalkBack/VoiceOver, haptic and visual feedback — nothing communicated by sound alone.

12

Built to grow

Admin panel for lessons, dictionary and sign mappings, and a consent-based recorder to grow the Egyptian sign dataset.

APPENDIX

Eshara feature inventory

All features extracted from the Eshara AI, Back-End, Front-End (Web), Mobile (Flutter) and Security requirements.

Sign → Text / Speech

- Live camera recognition (front/rear)
- On-device hand, pose & face landmarks
- Letters, words, phrases, sentences, continuous signing
- Running sentence area with confidence handling
- Arabic Text-to-Speech with mute & speed
- Offline letters & small word set

Text / Voice → Sign

- Arabic text input (RTL)
- Voice input → editable text
- Sequence player: play, pause, replay, speed, word highlight
- Pre-recorded sign clips with caching
- Fingerspelling for unmapped words
- 3D avatar: neutral, male, female

Live Conversation

- Split-screen two-way conversation on one phone
- Transcript with speaker labels & timestamps
- Flip mode for face-to-face reading

Dictionary

- Search in Arabic & English
- Sign detail: meaning, media, how to perform
- Browse by category
- Offline favorites
- Open entry from a translation result

Learning & Gamification

- Categories, lessons & lesson viewer
- Practice activities
- Camera practice with feedback
- Progress per category & lesson
- XP & streak
- Daily challenges
- Opt-in leaderboard with nicknames
- Mini games

Kids Mode

- Games & learning for children
- No chat, no leaderboard

Communication

- One-to-one chat (Firebase)
- Push notifications (FCM)
- Learning reminders
- Audio & video calls (WebRTC)
- Live captions / sign translation in calls

Emergency

- Egyptian numbers 122 / 123 / 180
- Visual emergency signs & responder cards
- Works fully offline
- One-tap call & home shortcut
- SMS with location to a saved contact

History & Account

- Save to history by choice
- Filter, view, delete, share
- Profile & avatar preference
- Light/dark, Arabic/English, text size, haptics
- Biometric unlock

Security & Privacy

- Raw video stays on device
- HTTPS / WSS only, JWT auth
- Users access only their own data
- Input validation
- Login-attempt blocking
- Secure token storage (Keychain / Keystore)

REFERENCES

Sources

Hand Talk — Google Play — https://play.google.com/store/apps/details?id=br.com.handtalk

Al-Turjuman — Tawasal product page — http://hayat.tawasal.org.sa/ar/product/altrjman-fy-lght-alashart

Al-Turjuman — Al-Araby report — www.alaraby.com (article)

Sign Language AI — Google Play — https://play.google.com/store/apps/details?id=ai.terp.www.twa

Lingvano — awards & streaks — www.lingvano.com (article)

Arabic Sign Language App for the Deaf — Google Play — https://play.google.com/store/apps/details?id=com.khokhar.yousaf.deafsignarabicapp

BrightSign Glove — https://www.brightsignglove.com/

PopSign, Ava, Hand Talk platform details — Eshara Survey (Part I)

Competitor features reflect their public store listings and product pages as of October 2026; a feature not mentioned there is marked ✗.